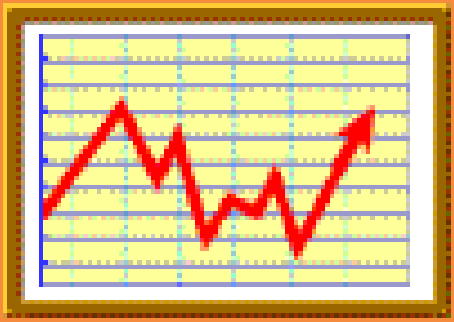
California State University, East Bay
College of Business and Economics

BAN 673 Time Series Analytics

Smoothing Methods: Exponential Smoothing

Lecture Materials



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Lecture Objectives

- **Define simple exponential smoothing (SES), its formula and smoothing constant**
- **Apply simple exponential smoothing in forecasting, and explain how to choose smoothing constant**
- **Identify and apply double exponential smoothing (Holt’s model) for series with additive and multiplicative trends**
- **Apply Winter’s (Holt-Winter’s) exponential smoothing for series with additive and multiplicative trend and seasonality**
- **Use R to develop smoothing methods for time series forecasting and for measuring forecast accuracy**

Types and of Exponential Smoothing

- ***Main types of exponential smoothing***
  - ***Simple exponential smoothing (SES)***
  - ***Advanced exponential smoothing (AES)***
    - » ***Holt's model*** **for data with trend**
    - » ***Winter's (Holt-Winter's) model*** **for data with trend and seasonality**
- ***Main principles of exponential smoothing***
  - ***Data-driven methods*** **– estimate time series components (level, trend, and seasonality) directly from the data without a predetermined structure**
  - ***Smooth out*** **the noise in a series in order to uncover the data patterns**
  - ***Average series values*** **over multiple time periods with various types of smoothers, i.e., applying smoothing constants for level, trend, and seasonality**

Simple Exponential Smoothing (SES)

- ***Simple exponential smoothing (SES)*** **is a popular forecasting methods.**
- **Popularity derived from the method's**
  - **Flexibility**
  - **Ease of automation**
  - **Cheap computation**
  - **Relatively good performance**
- **Similar to moving average in that it averages historical data periods to smooth out noise and uncover historical data patterns**
- **Major difference of simple exponential smoothing:**
  - **Take a *weighted average of all past historical data periods*, so that the weights decrease exponentially into the past**
  - **Give *more weight to recent historical data periods*, yet not completely ignore older historical data periods**
  - **Typically, should be used to forecasting series that has no trend and seasonality**

Simple Exponential Smoothing (SES) Forecast

- **Assumption:**
  - **Typically (but not always), series has only *level* (L_t) and *noise***
  - **Noise is unpredictable and level will "stay put"**
  - **Historical data is available up to period *t***
- **Forecast in period *t+1***

$$F_{t+1} = \alpha y_t + (1-\alpha)F_t = F_t + \alpha(y_t - F_t) = F_t + \alpha e_t$$

$$F_{t+k} = F_{t+1}$$

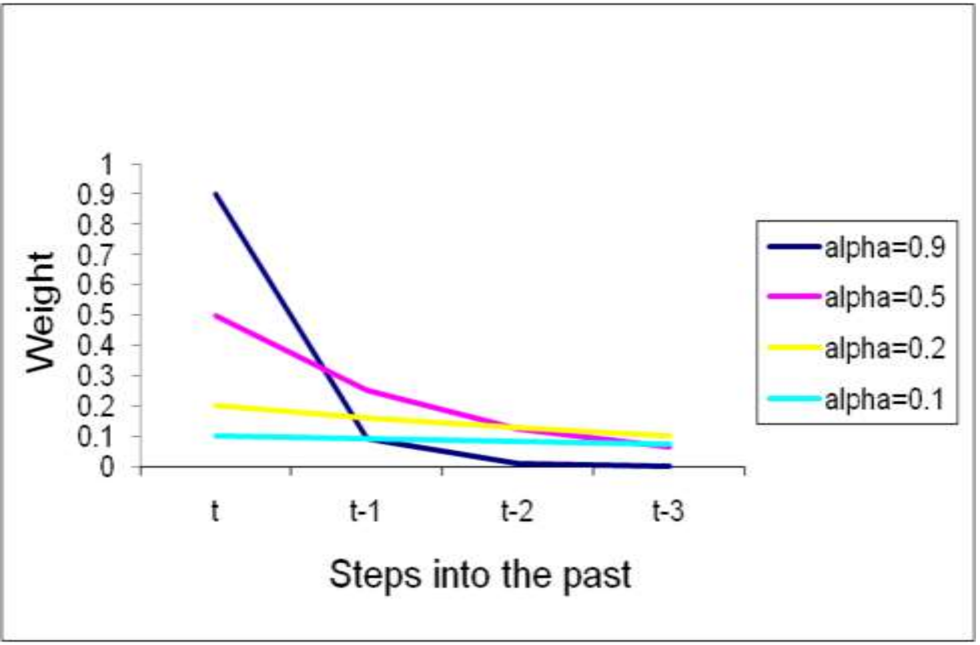
where α = ***smoothing constant*** **($0 \leq \alpha \leq 1$)**

- **Initialization for the first forecast: $F_1 = y_1$**
- **Why is it called "exponential smoothing"?**

$$F_{t+1} = \alpha y_t + \alpha(1-\alpha)\, y_{t-1} + \alpha(1-\alpha)^2 y_{t-2} + \alpha(1-\alpha)^3 y_{t-3} + \ldots\ \alpha(1-\alpha)^{n-1} y_1$$

- ***Weights decrease exponentially into the past!***

"Feeling" Effect of α

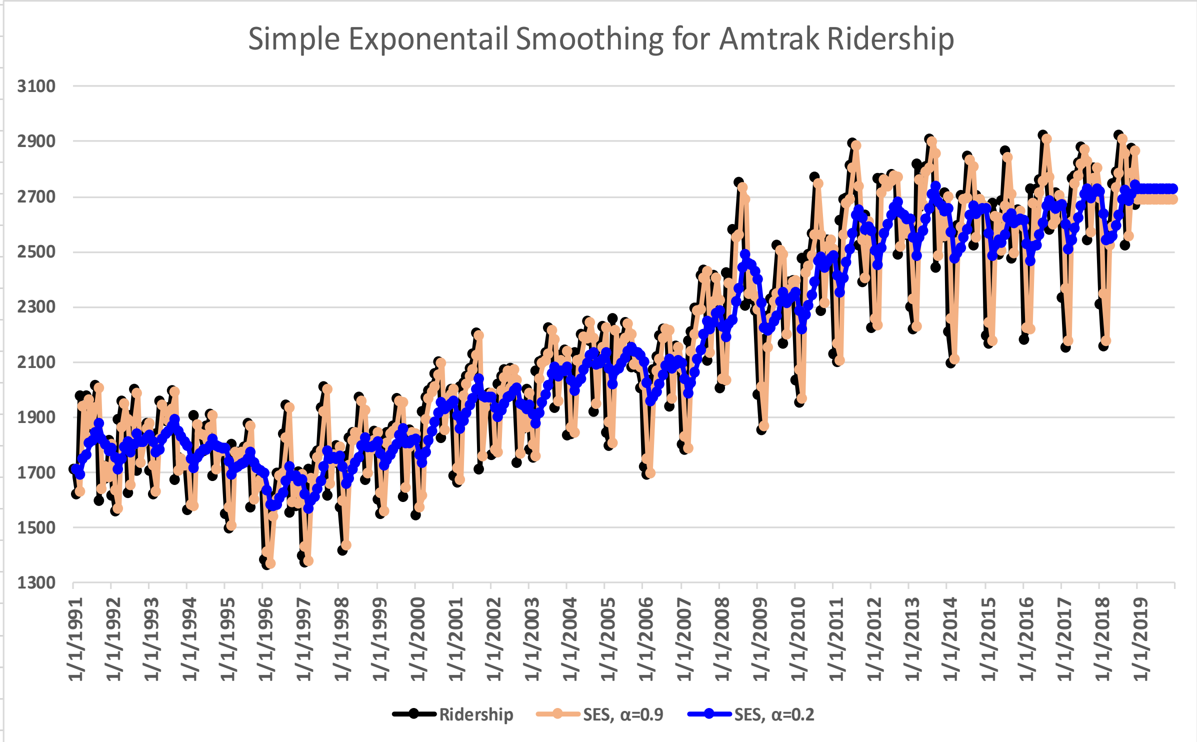


α	$\alpha(1-\alpha)$	$\alpha(1-\alpha)^2$	$\alpha(1-\alpha)^3$
0.9	0.09	0.009	0.0009
0.5	0.25	0.125	0.0625
0.2	0.16	0.128	0.1024
0.1	0.09	0.081	0.0729

$$F_{t+1} = \alpha y_t + \alpha(1-\alpha)\, y_{t-1} + \alpha(1-\alpha)^2 y_{t-2} + \alpha(1-\alpha)^3 y_{t-3} + \ldots$$

Example of SES in Excel

Month	Period	Ridership	SES, α=0.9	SES, α=0.5	SES, α=0.2
1/1/1991	1	1708.917	#N/A	#N/A	#N/A
2/1/1991	2	1620.586	1708.917	1708.917	1708.917
3/1/1991	3	1972.715	1629.419	1664.752	1691.251
4/1/1991	4	1811.665	1938.385	1818.733	1747.544
5/1/1991	5	1974.964	1824.337	1815.199	1760.368
6/1/1991	6	1862.356	1959.901	1895.082	1803.287
7/1/1991	7	1939.86	1872.111	1878.719	1815.101
8/1/1991	8	2013.264	1933.085	1909.289	1840.053
9/1/1991	9	1595.657	2005.246	1961.277	1874.695
10/1/1991	10	1724.924	1636.616	1778.467	1818.887
11/1/1991	11	1675.667	1716.093	1751.695	1800.095
12/1/1991	12	1813.863	1679.710	1713.681	1775.209
1/1/1992	13	1614.827	1800.448	1763.772	1782.940
2/1/1992	14	1557.088	1633.389	1689.300	1749.317
3/1/1992	15	1891.223	1564.718	1623.194	1710.871
4/1/1992	16	1955.981	1858.573	1757.208	1746.942
5/1/1992	17	1884.714	1946.240	1856.595	1788.750
6/1/1992	18	1623.042	1890.867	1870.654	1807.942
7/1/1992	19	1903.309	1649.824	1746.848	1770.962
8/1/1992	20	1996.712	1877.961	1825.079	1797.432
9/1/1992	21	1703.897	1984.837	1910.895	1837.288
10/1/1992	22	1810	1731.991	1807.396	1810.610
11/1/1992	23	1861.601	1802.199	1808.698	1810.488
12/1/1992	24	1875.122	1855.661	1835.150	1820.710



SES, α = 0.2, at 3/1991, t=3:

SES$_3$: $F_3 = \alpha y_2 + (1-\alpha)F_2$ ***= 0.2*1620.586 + (1-0.2)*1708.917 = 1691.251***

SES, α = 0.2, at 6/1991, t=6:

SES$_6$: $F_6 = \alpha y_5 + (1-\alpha)F_5$ ***= 0.2*1974.964 + (1-0.2)*1760.368 = 1803.287***

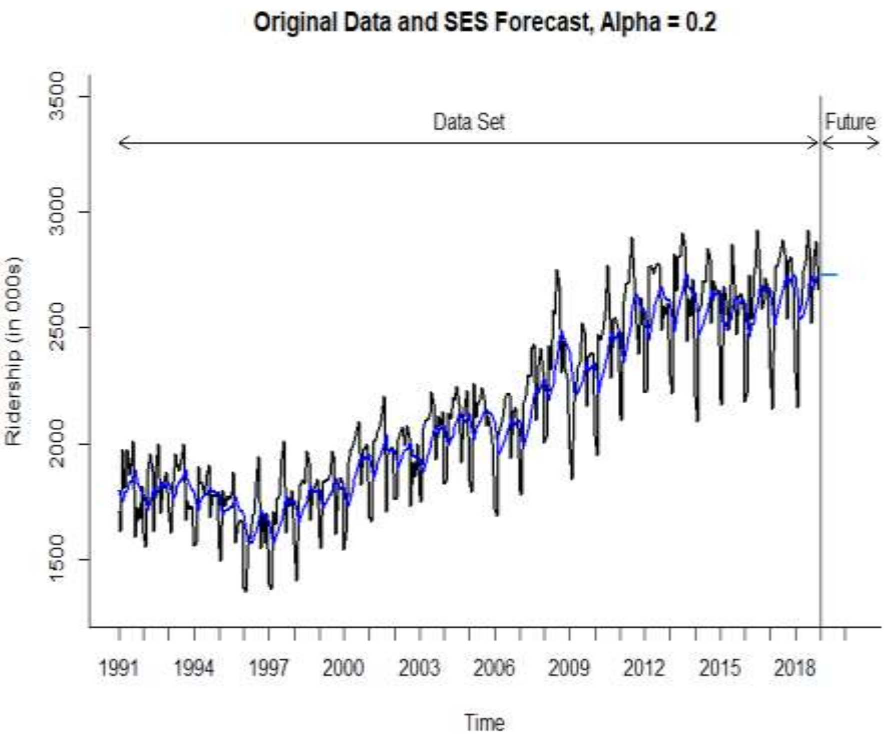
SES, α = 0.2, at 1/2019, t=337:

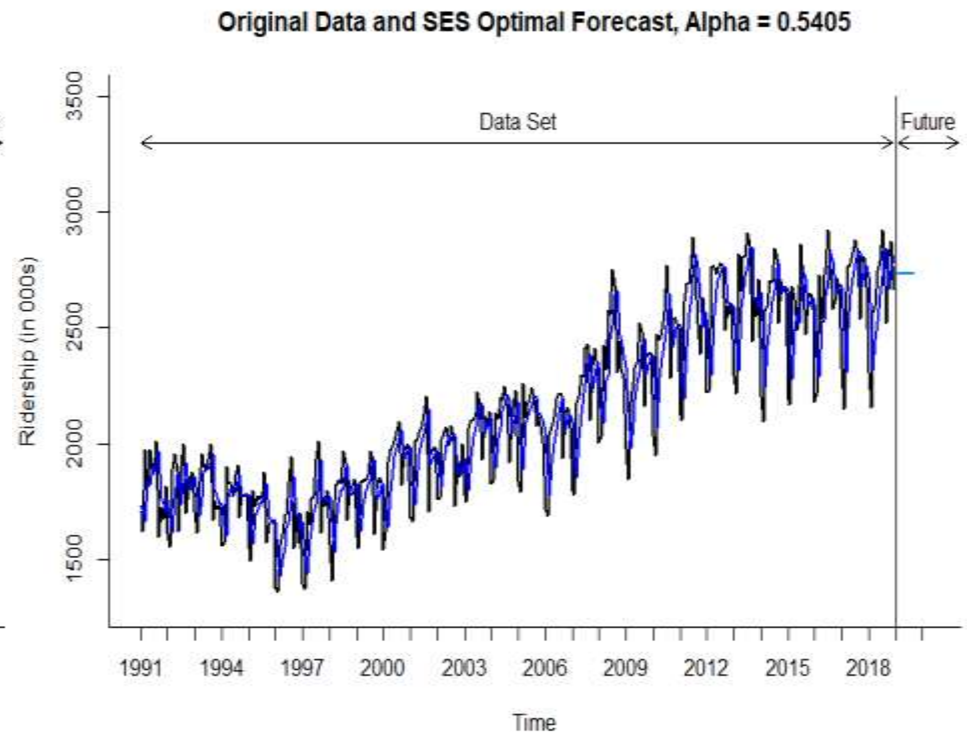
SES$_{337}$: $F_{337} = \alpha y_{336} + (1-\alpha)F_{336}$ ***= 0.2*2668.049 + (1-0.2)*2740.186 = 2725.759, will be used for all periods in 2019 because no actual data is available in 2019***

Choosing Smoothing Constant α

- **Smoothing constant determines the *rate of learning* and is *set by user***
  - ***$0 \leq \alpha \leq 1$***
  - ***$\alpha \to 1$*** **indicates fast learning, i.e., only the most recent values influence on the forecast – *under-smoothing***
  - ***$\alpha \to 0$*** **indicates slow learning, i.e., past observations have large influence on the forecast – *over-smoothing***
- **Choice of α depends on the required level of smoothing and on how relevant the history is for generating forecasts**
  - **Common values α = *0.1 - 0.2***
  - **Trial and error with various α values can also help; plot and visualize various SES charts**
  - **Apply performance/accuracy measure like MAPE and RMSE to compare validation forecasts; using this just for training data can lead to model overfitting**
- **In R, a forecast using simple exponential smoothing can be done using *ets()* function, which stands for *error, trend, and seasonality*, respectively**
  - **Utilize *model* = *"ANN"*, where *A* = additive error, first *N* = no trend, second *N* = no seasonality**
  - **Apply a specific α value, e.g., α = *0.2***
  - **If you choose to find an optimal value of α, then don't apply α in the *ets()* function**

Visualizing Two SES Forecasts for Amtrak Ridership





SES, α = 0.2

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	13.823	185.696	154.581	-0.078	7.415	0.357	0.936

Optimal SES, α = 0.5405

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	5.524	183.986	146.196	-0.313	7.039	0.14	0.941

Advanced Exponential Smoothing: Holt's Model

- ***Holt's (double-exponential) smoothing*** **is used for time series that contains trend**
  - **Idea is to augment simple exponential smoothing (SES) by capturing a trend component**
- ***Holt's additive model:*** ***Forecast = Level (L_t) + Trend(T_t)***

$$F_{t+k} = L_t + k\, T_t$$

- ***Level component (L_t)*** **(exponential smoothing)**

$$L_t = \alpha y_t + (1-\alpha)(L_{t-1} + T_{t-1})$$

- ***Trend component (T_t)*** **(linear trend)**

$$T_t = \beta\,(L_t - L_{t-1}) + (1- \beta)\, T_{t-1}$$

where: α **= smoothing constant for exponential smoothing ($0 \le \alpha \le 1$)**

β **= smoothing constant for trend estimate ($0 \le \beta \le 1$)**

k **= periods to be forecasted into future**

- **Choosing α and β**
  - **Default values, e.g., $\alpha, \beta = 0.1 – 0.2$**
  - **Compare several Holt's models with various α and β to identify such α and β that would minimize *RMSE* or *MAPE* for training set**
    - » **Overfitting may be a problem**
    - » **What to do: make sure chosen values are reasonable**

Example of Holt's Model in Excel

Month	Period	Ridership	L_t	T_t	F_{t+1}
1/1/1991	1	1708.917	**1708.917**	**0.000**	#N/A
2/1/1991	2	1620.586	1700.084	-0.883	1708.917
3/1/1991	3	1972.715	1726.552	1.852	1699.201
4/1/1991	4	1811.665	1736.730	2.684	1728.404
5/1/1991	5	1974.964	1762.969	5.040	1739.414
6/1/1991	6	1862.356	1777.444	5.983	1768.009
7/1/1991	7	1939.86	1799.071	7.548	1783.427
8/1/1991	8	2013.264	1827.283	9.614	1806.618
9/1/1991	9	1595.657	1812.773	7.202	1836.897
10/1/1991	10	1724.924	1810.470	6.251	1819.975
11/1/1991	11	1675.667	1802.616	4.841	1816.721
12/1/1991	12	1813.863	1808.097	4.905	1807.456
1/1/1992	13	1614.827	1793.184	2.923	1813.002
2/1/1992	14	1557.088	1772.206	0.533	1796.107
3/1/1992	15	1891.223	1784.587	1.718	1772.738
4/1/1992	16	1955.981	1803.272	3.414	1786.305
5/1/1992	17	1884.714	1814.489	4.195	1806.687
6/1/1992	18	1623.042	1799.120	2.238	1818.684
7/1/1992	19	1903.309	1811.553	3.258	1801.358
8/1/1992	20	1996.712	1833.001	5.077	1814.811
9/1/1992	21	1703.897	1824.660	3.735	1838.078
10/1/1992	22	1810	1826.555	3.551	1828.395
11/1/1992	23	1861.601	1833.256	3.866	1830.107
12/1/1992	24	1875.122	1840.922	4.246	1837.122
1/1/1993	25	1705.259	1831.177	2.847	1845.168
2/1/1993	26	1618.535	1812.475	0.692	1834.024
3/1/1993	27	1836.709	1815.521	0.927	1813.167
4/1/1993	28	1957.043	1830.508	2.333	1816.449
5/1/1993	29	1917.185	1841.276	3.177	1832.842

α	β
0.10	0.10

Holt's Additive Model

$L_t = \alpha y_t + (1-\alpha)(L_{t-1} + T_{t-1})$

$T_t = \beta (L_t - L_{t-1}) + (1-\beta) T_{t-1}$

$F_{t+1} = L_t + T_t$

Intial (seed) parameters for the Holt't model in period 1:

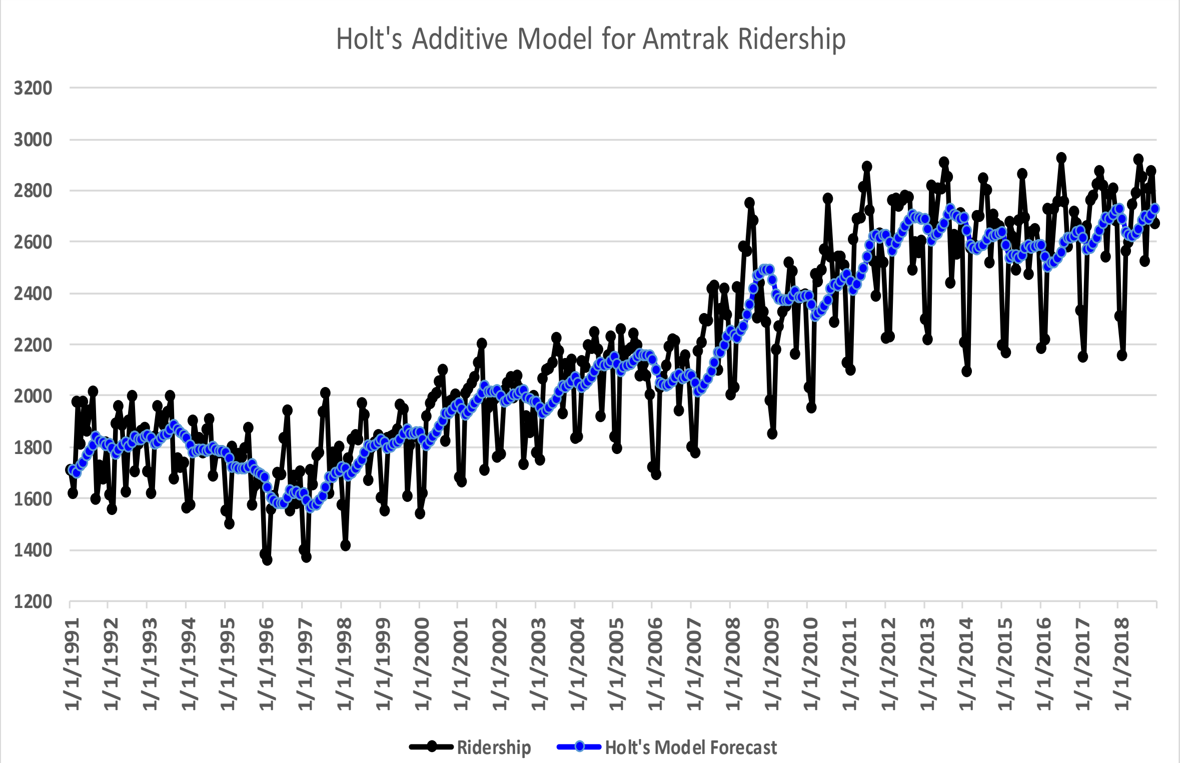
L_t = 1708.917, the actual value for period 1

T_t = 0, commonly defined trend in period 1

$L_2 = 0.1*1620.586 + (1-0.1)*(1708.917 + 0.000) = 1700.084$

$T_2 = 0.1*(1700.084 - 1708.917) + (1-0.1)*0.000 = -0.883$

$F_3 = 1700.084 - 0.883 = 1699.201$



Holt's Model (continued)

- ***Holt's multiplicative model:*** *Forecast = Level (L_t) × Trend(T_t)*

$$F_{t+k} = L_t \times T_t^k$$

$$L_t = \alpha y_t + (1-\alpha)(L_{t-1} \times T_{t-1})$$

$$T_t = \beta (L_t / L_{t-1}) + (1-\beta)\, T_{t-1}$$

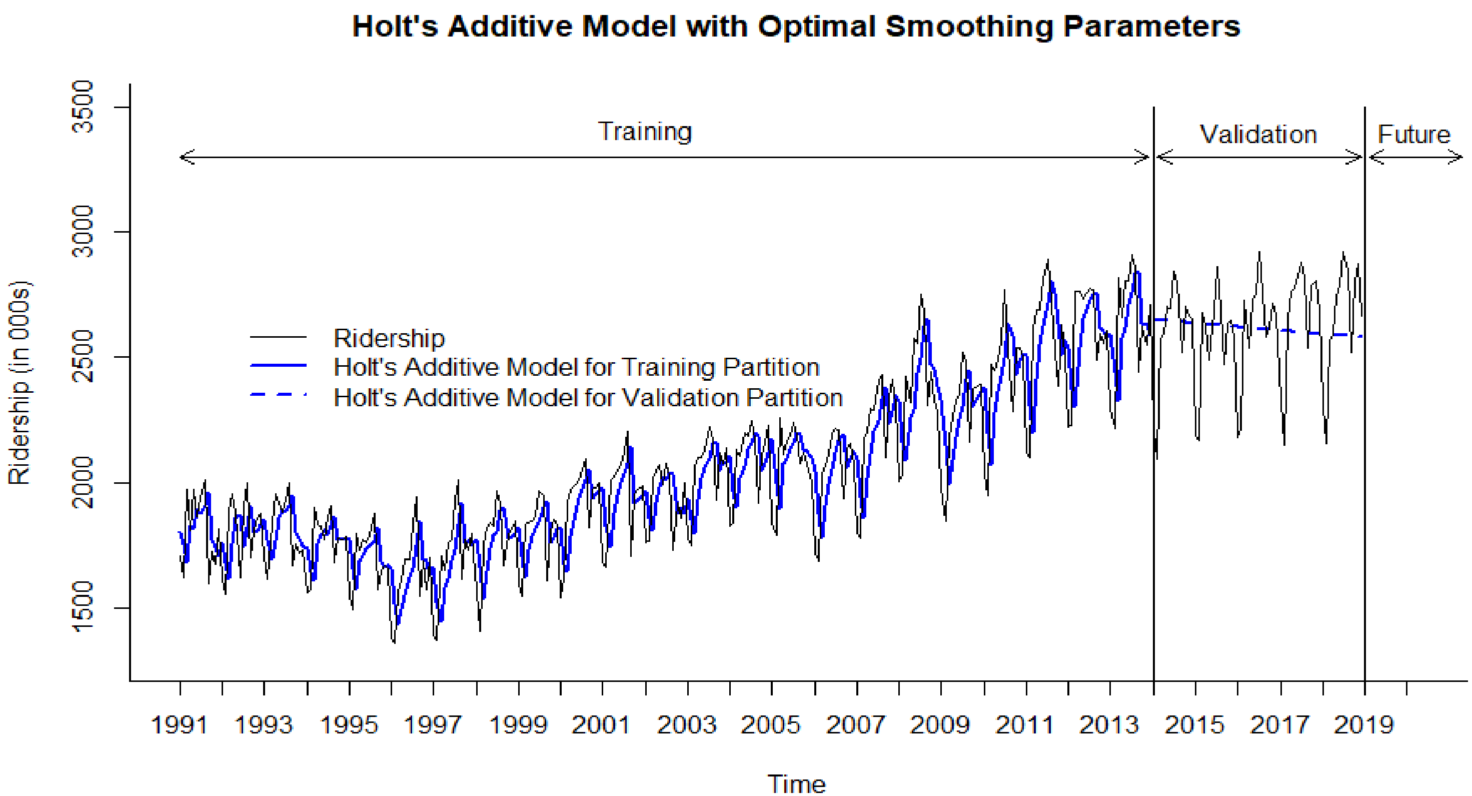
where: α = smoothing constant for exponential smoothing ($0 \le \alpha \le 1$)

β = smoothing constant for trend estimate ($0 \le \beta \le 1$)

k = periods to be forecasted into future

- In R, to automatically select Holt's model, use *model = "ZZN" in ets()* function
  - First *Z* can be *A* (additive) or *M* (multiplicative) error
  - Second *Z* can be *A* (additive) or *M* (multiplicative) trend
  - Third *N* means the Holt's model, which does not have seasonality parameter
  - *Example: model = "AMN"*, where *A* = additive error, *M* = multiplicative trend, and *N* = no seasonality
- For automatic selection of α and β, don't apply these parameters in the *ets()* function

Holt's Additive Model (AAN) with Optimal Smoothing Parameters



Molel (A, A, N)
Smoothing Parameters:
$\alpha = 0.5118$
$\beta = 0.0001$

Advanced Exponential Smoothing: Holt-Winter's Model

- ***Holt-Winter's (HW) or simply Winter's model*** **is used for time series that contains trend and seasonality**
  - **Idea is to augment *Holt's model* by capturing a seasonal component**
- ***Winter's multiplicative model:*** ***Forecast = [Level (Lt) + Trend(Tt)]×Seasonal Component (S_t)***

$$F_{t+k} = (L_t + k\,T_t) \times S_{t+k-M}$$

where: S_{t+k-M} **= seasonal index of period *t+k-M* (*M* = number of seasons)**

- ***Level component (L_t)*** **(exponential smoothing)**

$$L_t = \alpha \frac{y_t}{S_{t-M}} + (1-\alpha)(L_{t-1} + T_{t-1})$$

- ***Trend component (T_t)*** **(linear trend, same as Holt's model)**

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)\,T_{t-1}$$

- ***Seasonal component (S_t)*** **(multiplicative)**

$$S_t = \gamma \frac{y_t}{L_t} + (1-\gamma)\,S_{t-M}$$

where:

α **= smoothing constant for exponential smoothing ($0 \le \alpha \le 1$)**

β **= smoothing constant for trend estimate ($0 \le \beta \le 1$)**

γ **= smoothing constant for seasonality estimate ($0 \le \gamma \le 1$)**

k **= periods to be forecasted into future**

M **= number of seasons**

Holt-Winter's Model (continued)

- ***Holt-Winter's additive model:*** *Forecast = Level (Lt) + Trend(Tt) + Seasonal Component (S_t)*

$$F_{t+k} = L_t + k\,T_t + S_{t+k-M}$$

$$L_t = \alpha(y_t - S_{t-M}) + (1-\alpha)(L_{t-1} + T_{t-1})$$

$$T_t = \beta(L_t - L_{t-1}) + (1-\beta)\,T_{t-1}$$

$$S_t = \gamma(y_t - L_t) + (1-\gamma)\,S_{t-M}$$

where: S_{t+k-M} = seasonal additive component for period $t+k-M$ (M = number of seasons)

α = smoothing constant for exponential smoothing ($0 \le \alpha \le 1$)

β = smoothing constant for trend estimate ($0 \le \beta \le 1$)

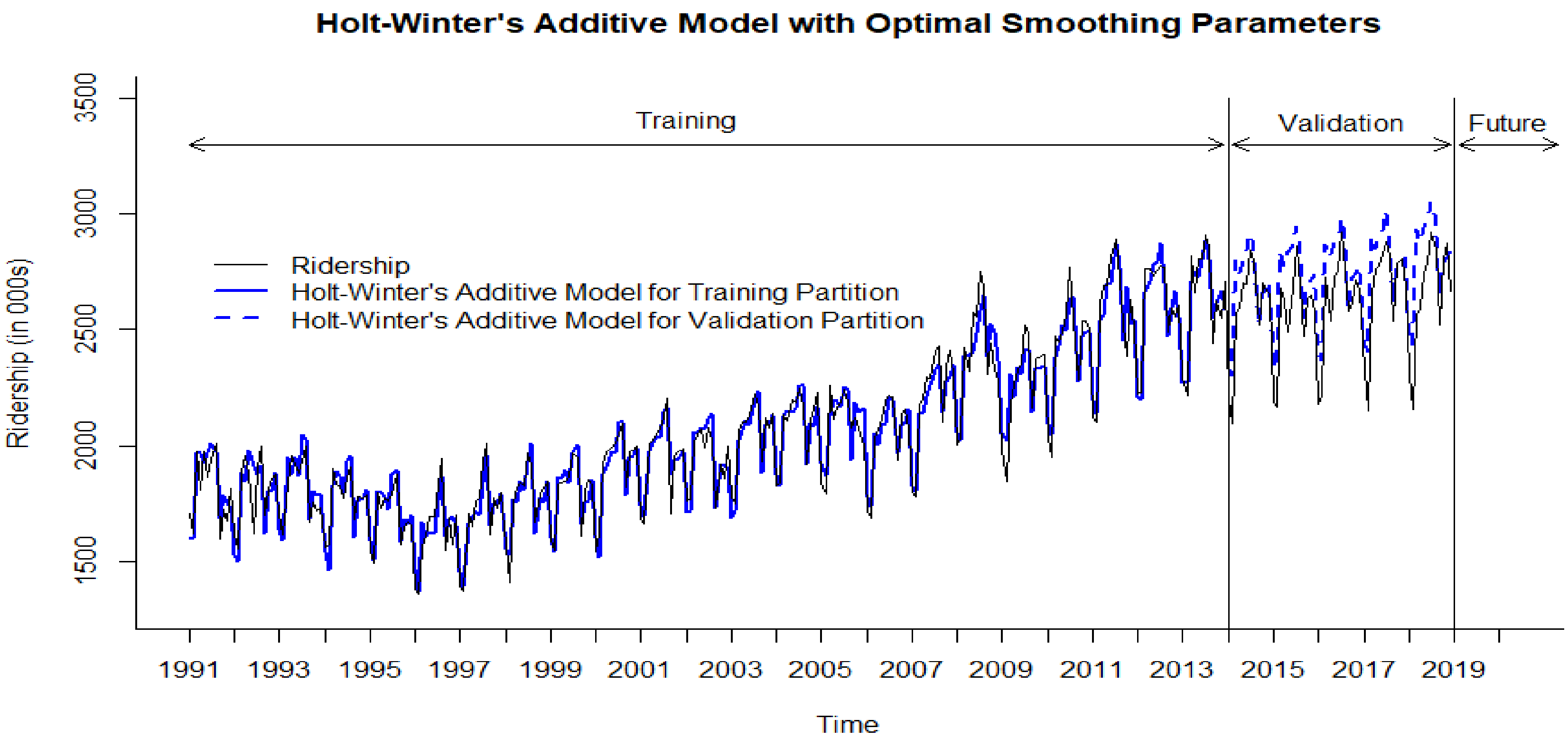
γ = smoothing constant for seasonality estimate ($0 \le \gamma \le 1$)

k = time periods to be forecasted into future

M = number of seasons

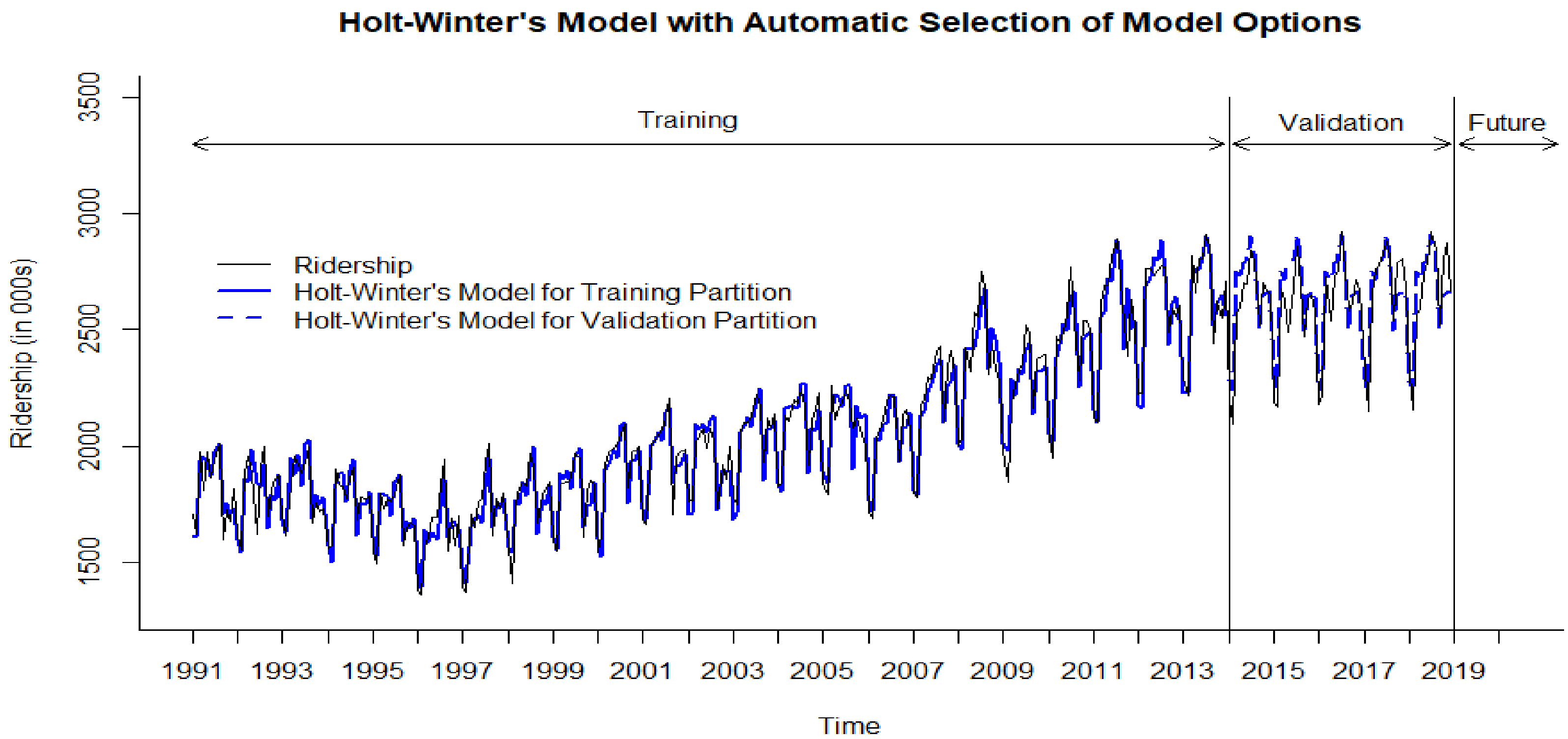
- **In R, to automatically select Holt-Winter's model, use *model = "ZZZ" in ets()* function**
  - **First *Z* can be equal to *A* (additive) or *M* (multiplicative) error**
  - **Second *Z* can be equal to *A* (additive) or *M* (multiplicative) trend**
  - **Third Z can be equal to *A* (additive) or *M* (multiplicative) seasonality**
  - ***Example: model = "MAA"*, where *M* = multiplicative error, *A* = additive trend, and *A* = additive seasonality; it is a *Holt-Winter's additive model with multiplicative error***
- **For automatic selection of α, β and γ, don't apply these parameters in the *ets()* function**

Holt-Winter's Additive Model (AAA) with Optimal Smoothing Parameters



Molel (A, A, A)
Smoothing Parameters:
$\alpha = 0.4792$
$\beta = 0.0001$
$\gamma = 0.2261$

Holt-Winter's Model - Automated Selection of Model Options (ZZZ) and Optimal Parameters



Molel (M, N, M)
Smoothing Parameters:
$\alpha = 0.5156$
$\beta = 0$ (not used)
$\gamma = 0.158$

Performance Measures of SES and Holt-Winter's Models Using Accuracy Function in R

SES, α = 0.2

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-40.446	218.813	159.056	-2.304	6.513	0.456	0.955

HW Additive Model with Optimal Parameters

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-127.422	155.824	132.872	-5.039	5.233	0.534	0.696

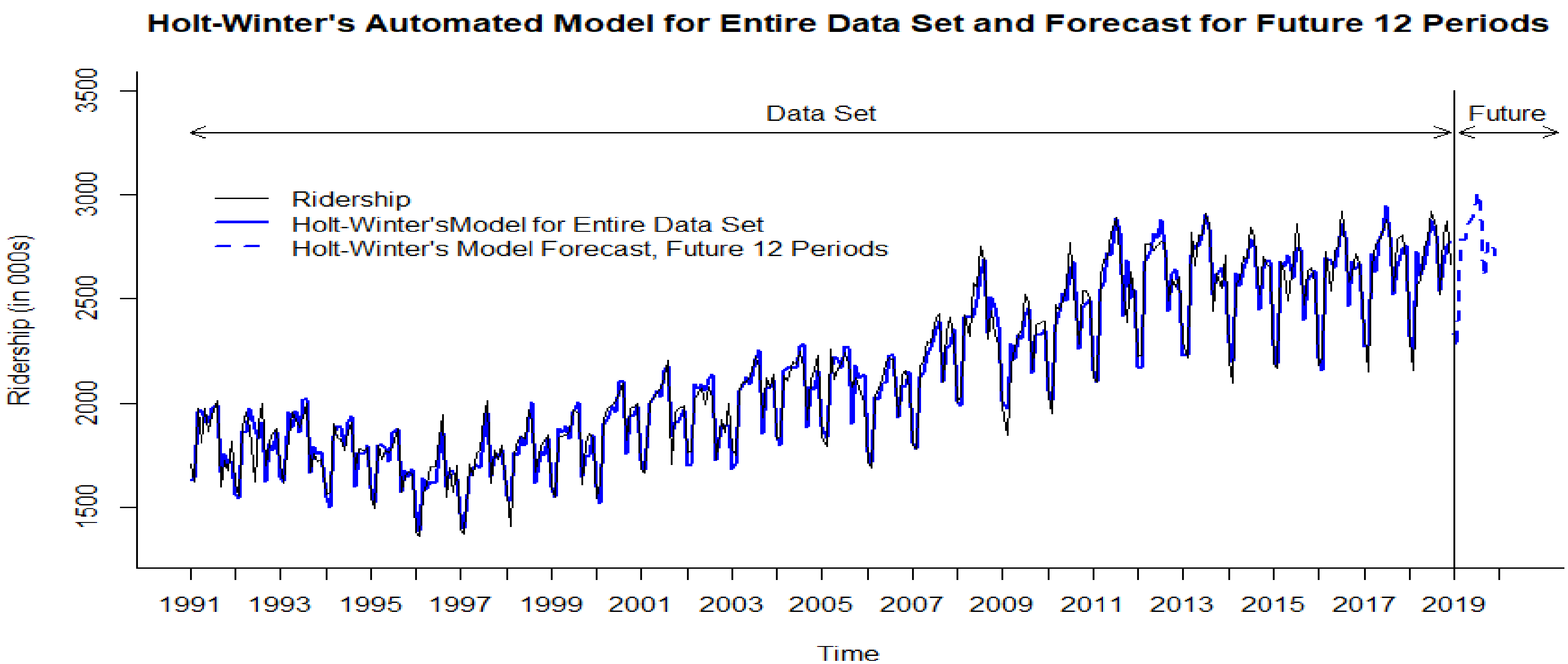
HW Model with Automatic Selection of Model Options

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	-33.28	93.659	71.321	-1.389	2.781	0.546	0.41

Forecasting for Future Periods

- **Before attempting to forecast future values of time series, *the training and validation periods must be recombined into one (entire) time series data set***
- **The chosen model, or several models, e.g., *HW model with automatic selection of model options* and *HW additive model (AAA) with optimal parameters*, *need to be rerun on the entire data set***
- ***Measure model or several models' performance vs. original data* using accuracy measures like *MAPE*, *RMSE* and others**
- **The new forecasting model, based on the entire data set, can be now used for forecasting future periods**

Holt-Winter's Model with Automatic Selection of Model Options and Forecast for Future Periods



Molel (M, Ad, M)
Smoothing Parameters:
$\alpha = 0.5334$
$\beta = 0.0014$
$\gamma = 0.1441$
$\phi = 0.9698$

Damped Holt's and Holt-Winter's Model

- **The forecasts generated by Holt's or Holt-Winter's model display a constant trend (increasing or decreasing) indefinitely into the future**
- **Empirical evidence indicates that these methods tend to *over-forecast*, especially for *longer forecast horizons***
- **Gardner & McKenzie* introduced a *damping parameter* ϕ *(phi)*, $0 < \phi < 1$, that "dampens" the trend to a flat line sometime in the future**
  - **If $\phi = 1$, the model is identical to the regular Holt's or Holt-Winter's model**
  - **For values between 0 and 1, ϕ dampens the trend so that it approaches a constant sometime in the future**
  - **In practice, ϕ is rarely less than 0.8 as the damping has a very strong effect for smaller values. In *R*, the *Forecast* package restricts ϕ to a minimum of 0.8 and a maximum of 0.98**
- **Holt's and Holt-Winter's models that include a damped trend parameter ϕ have proven to be very successful and are arguably the most popular individual methods when forecasts are required automatically for many series**

*** From R.J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, https://otexts.com/fpp2/holt.html and https://otexts.com/fpp2/holt-winters.html. Retrieved on December 5, 2022**

Performance Measures of Holt-Winter's Models and Seasonal Naïve Model for Entire Data Set

HW Model with Automatic Selection of Model Options

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	5.022	68.419	54.027	0.147	2.549	-0.022	0.342

HW Additive Model with Optimal Parameters

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	0.078	72.936	57.721	-0.12	2.743	0.127	0.368

Seasonal Naïve Model

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	31.113	116.076	93.101	1.26	4.375	0.631	0.579